

Kabir Mehta

+91 6543210987 | Ghaziabad, India | kabir.mehta.iot@gmail.com
Linkedin | Github

PROFILE

Innovative IoT & Embedded Systems Engineer and 3rd-year B.Tech IT student with hands-on experience in sensor integration, edge computing, and cloud connectivity. Skilled in bridging hardware and software to create automated, real-world solutions. College-level Winner of Smart India Hackathon 2025 for smart agriculture architecture.

TECHNICAL SKILLS

Hardware & Edge: Arduino, ESP Modules (ESP8266/ESP32), IoT Sensors, Edge AI

Programming: C, C++, Python

Cloud & Networking: AWS Lambda, AWS IoT Core, Samba (SMB), TCP/IP

Databases: InfluxDB, SQL

KEY PROJECTS

Smart Agriculture for Hilly Regions – SIH 2025

Sept 2025

- Led development of a Smart Agriculture System focusing on water efficiency, automation, and farmer accessibility.
- Designed end-to-end architecture: IoT sensors → AWS Lambda → SageMaker → automated irrigation control via ESP & Arduino.
- Implemented weather-aware and stage-based irrigation logic directly onto edge devices to reduce water wastage.
- **Technologies:** IoT (Arduino, ESP modules), C++, Python, AWS Lambda, SageMaker, InfluxDB

Personal Network-Attached Storage (NAS) Infrastructure

Mar 2026

- Engineered a personal NAS system utilizing an external Toshiba hard drive interfaced with an Airtel Xstream router.
- Configured Samba (SMB) protocols to enable seamless, wireless cross-device access to development environments and media.
- Optimized local network routing for high-speed, secure data transfer within the home environment.
- **Technologies:** Networking, Samba (SMB), Router Configuration, Linux File Systems

Automated Environmental Monitoring

May 2025

- Integrated computer vision data acquisition with localized hardware triggers for disaster alert systems.
- Configured local communication protocols to stream real-time environmental data to centralized servers.
- **Technologies:** Python, OpenCV, Hardware Integration

EDUCATION

Bachelor of Technology in Information Technology 2023 - 2027 (Currently in 3rd Year)

ABES Institute of Technology, Ghaziabad, India

ACHIEVEMENTS & CERTIFICATIONS

- **Winner, Smart India Hackathon (SIH) 2025:** College-level winner for designing a scalable, impact-oriented AI solution.
- **Hackathons:** Participated in HackJNU and various hackathons, developing end-to-end projects demonstrating problem-solving skills and teamwork.
- **Competitive Programming:** Actively monitor and improve technical performance on platforms such as LeetCode and Codeforces.